*A fully edible, soft, pneumatic actuator undergoing a gripper grasping test (Credit: EPFL)*

organisation EPFL has made some headway recently. At a conference held this year, researchers led by Dario Floreano presented the prototype of a completely edible, soft, pneumatic actuator made of gelatin, glycerin and water. The design and performance of this new gelatin actuator is comparable to standard pneumatic actuators. Its structure causes it to bend when inflated and straighten out again when pressure is reduced. The main benefits are that it is edible, biodegradable, biocompatible and environmentally sustainable. Since gelatin is melty, the actuator also turns out to be self-healing!

The researchers explain several exciting applications for this actuator. The components of such edible robots could be mixed with nutrients or pharmaceutical components to improve healing, digestion and metabolism. These can be used as disposable robots to explore, study the behaviour of wild animals, cure sick animals or train protected animals to hunt. These can also be used in relief measures. In search-and-rescue operations, the robot can be sent without a payload to stranded people as the robot itself is food!

## But can we pay the bills?

A recent Frost & Sullivan report noted that the fastest growing segment amongst the many types of implants is implantable neuro-stimulators, which help treat neurological disorders such as epilepsy, dementia, Alzheimer's disease, Parkinson's disease and dystonia. It is also likely that in the future, these implantable electronics will be digitally connected to improve the scope of remote drug delivery, testing and diagnostics.

The cost of researching and developing implants continues to be high due to the criticality and complications involved—and often this cost reflects in the end price points. In the report, Frost & Sullivan industry analyst Bhargav Rajan noted that the constant stream of innovations has attracted substantial private funding to the implantable electronics market, while public funding is expected to improve in the future. He also suggested that technology developers can lower development costs by collaborating with early-stage start-ups and small- and medium-sized enterprises. This will allow them access to cross-industry expertise and cutting-edge innovations, which, in turn, will help lower the price points.

A majority of the population seeking implants belongs to low- and middle-income groups. Taking note of this factor, the report also stressed that the growth of this sector depends not entirely on technological development but also the availability of insurance coverage and reimbursements for such devices. EFY